

We have a block sliding horizontally on a table (representing the wall) being pulled by the weight of a hanging block.

We can draw the forces acting on the two blocks. Note that they are all pointing along either the x or y axes. N_M is the normal force from the wall up on the heavy block, W_M is the heavy block's weight, T is the tension force on the rope, F_k is the force of kinetic friction *opposite* the direction of motion, and finally W_m is the weight of the light block. Recall that the tension force should pull towards the center of the rope.

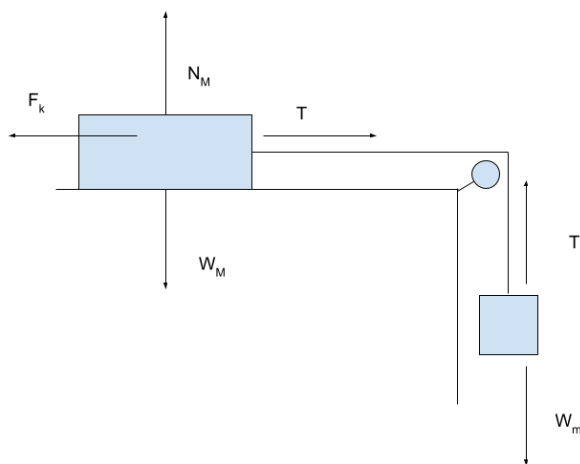


Figure 1: image

We can then set up the equations for Newton's second law in the x and y directions, first for the large block:

$$\begin{aligned}
 M : \\
 x : \quad & F_{NET,M}^x = -F_k + T & = Ma_M^x & (1) \\
 y : \quad & F_{NET,M}^y = -W_M + N_M & = Ma_M^y & (2) \\
 & & & (3)
 \end{aligned}$$

Note that F_k and W_M are negative, since they point in the $-x$ and $-y$ directions, respectively.

We know that the force of gravity on an object is given by $W_m = m * g$ where g is the constant representing the strength of gravity. Note that while g has units of acceleration, it really just tells us how fast an object *would* accelerate if in free fall. Objects that are stationary will still experience a gravitational force of mg - it must just have another force canceling it out, to ensure its vertical acceleration is 0. This is exactly what happens here, and is what the normal force does.

The block resting on the table is experiencing a weight force, but the normal force balances it out exactly (since the block is sitting on top of the wall, and obviously can't go down any more through the wall- it's being held up). Thus we have $a_M^y = 0$, so the right hand side of the y equation becomes $Ma_M^y = M * 0 = 0$

Thus the y equation becomes

$$\begin{aligned} F_{NET,M}^y &= -W_M + N_M = -Mg + N_M = Ma_M^y = 0 \\ -Mg + N_M &= 0 \\ N_M &= Mg \end{aligned} \quad (4)$$

Now let's look at the equations for the smaller, hanging mass m . Note that because there are no forces and no acceleration in the x direction for the hanging block, I only need to worry about the y equation, since the x equation is just $F_{NET,m}^x = 0 = ma_m^x = m * 0 = 0 \implies 0 = 0$ which says nothing new or interesting. Then we get

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$$\begin{aligned} m : \\ y : \end{aligned} \quad F_{NET,m}^y = T - mg = ma_m^y \quad (5)$$

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Now we note that a_m^y need not be 0. In fact it *shouldn't* be 0. Recall of a taut rope, that the motion of the two objects attached by the rope should be in unison. If we want the big block to be sliding and potentially accelerating towards the edge of the wall with the pulley (letting gravity do the work) then the acceleration of the horizontally to the right requires that the rope be pulled *down* by the hanging block. This means that the acceleration in the $-\hat{y}$ direction for the hanging block should be equal to the acceleration to the right of the sliding block. This can be quantified as

$$a_M^x = -a_m^y \equiv a$$

I use the minus sign to indicate that the positive acceleration in the x direction for the block M corresponds to *negative* acceleration in the y direction for block m . $\equiv a$ means I am defining a new variable a equal to both of a_M^x and $-a_m^y$.

Since we already solved the y equation for M and the x equation for m , we are left with two equations to solve.

$$F_{NET,M}^x = -\mu Mg + T = Ma_M^x \quad (6)$$

$$F_{NET,m}^y = T - mg = ma_m^y \quad (7)$$

We can substitute $a_M^x = -a_m^y \equiv a$.

$$F_{NET,M}^x = -F_k + T = Ma \quad (8)$$

$$F_{NET,m}^y = T - mg = -ma \quad (9)$$

From here there are two ways we can go about solving this. First, let's try substitution. Take the first of the above equations, and expand the kinetic friction using $F_k = \mu N_M$.

$$F_{NET,M}^x = -\mu Mg + T = Ma \quad (10)$$

$$T = Ma + \mu Mg \quad (11)$$

$$(12)$$

We can now plug this in to $F_{NET,m}^y$

$$F_{NET,m}^y = (Ma + \mu Mg) - mg = -ma \quad (13)$$

$$F_{NET,m}^y = -Ma - \mu Mg + mg = ma \quad (14)$$

$$mg - \mu Mg = ma + Ma \quad (15)$$

$$mg - \mu Mg = (m + M)a \quad (16)$$

$$(17)$$

We see that the left side has two forces, and the right hand side looks like the right side of Newton's Second Law, ie. $F = ma$. Because these two bodies are rigidly connected, we can treat them as one system, and rewrite the equation as

$$F_{NET,System} = M_{System}a_{System}$$

Where the net force on the system is $F_{NET,System} = mg - \mu Mg$, the combined mass of the system is $M_{System} = (m + M)$. Finally, since the system is rigid, the acceleration of the system is the acceleration of the sliding block and the acceleration of the hanging block, which are of the same magnitude.

One other way we can solve this is by immediately taking $F_{NET,m}^y = T - mg = -ma$ and multiplying by -1

$$F_{NET,m}^y = T - mg = -ma \quad (18)$$

$$= -T + mg = ma \quad (19)$$

$$(20)$$

If we now add this to $F_{NET,M}^x = -\mu Mg + T = Ma$

$$F_{NET,m}^y = -T + mg = ma \quad (21)$$

$$+ \quad + \quad (22)$$

$$F_{NET,M}^x = -\mu Mg + T = Ma \quad (23)$$

$$= \quad (24)$$

$$F_{NET,system} = mg - \mu Mg = (m + M)a \quad (25)$$